

Total No. of Questions : 10]

SEAT No. :

P2366

[Total No. of Pages : 2

[5870]-1534

T.E. (Computer Engineering)
Software Modeling and Design
(2015 Pattern) (Semester - II) (310253)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.
- 4) Figures to the right indicate full marks.

Q1) a) Define software design and give its importance. [5]

b) Define component. Compare component and deployment diagram. [5]

OR

Q2) a) Explain 4 + 1 View Architecture of UML. [5]

b) Draw Use Case diagram for ATM system by considering cash withdrawal scenario. [5]

Q3) a) Draw a sequence diagram for Online Shoppe. System. [5]

b) Define interface. Explain required and provided interface with example. [5]

OR

Q4) a) Explain with an example the difference between aggregation and composition. [5]

b) Draw Class Diagram for College Library Management System. [5]

Q5) a) Explain synchronous communication pattern in client server architecture. [8]

b) Explain the broker pattern for service oriented architecture. [8]

OR

Q6) a) Explain the component based software architecture in detail. [8]

b) Explain the real time software architecture. [8]

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- Q7)** a) Explain factory pattern with its intent, motivation and implementation. [8]
b) Explain observer pattern with suitable class diagram including subject and observer. [8]

OR

- Q8)** a) Explain proxy pattern with its intent, motivation and implementation. [8]
b) Explain Behavioral Design Pattern with example. [8]

- Q9)** a) Why testing is important? List different testing methods. [6]
b) What is integration testing? Explain its types in details. [6]
c) What is system testing? Explain with different types of testing. [6]

OR

- Q10)** a) Write a short note on (Any 3); [18]
i) Performance testing
ii) Scenario testing
iii) Unit testing
iv) Acceptance testing.

